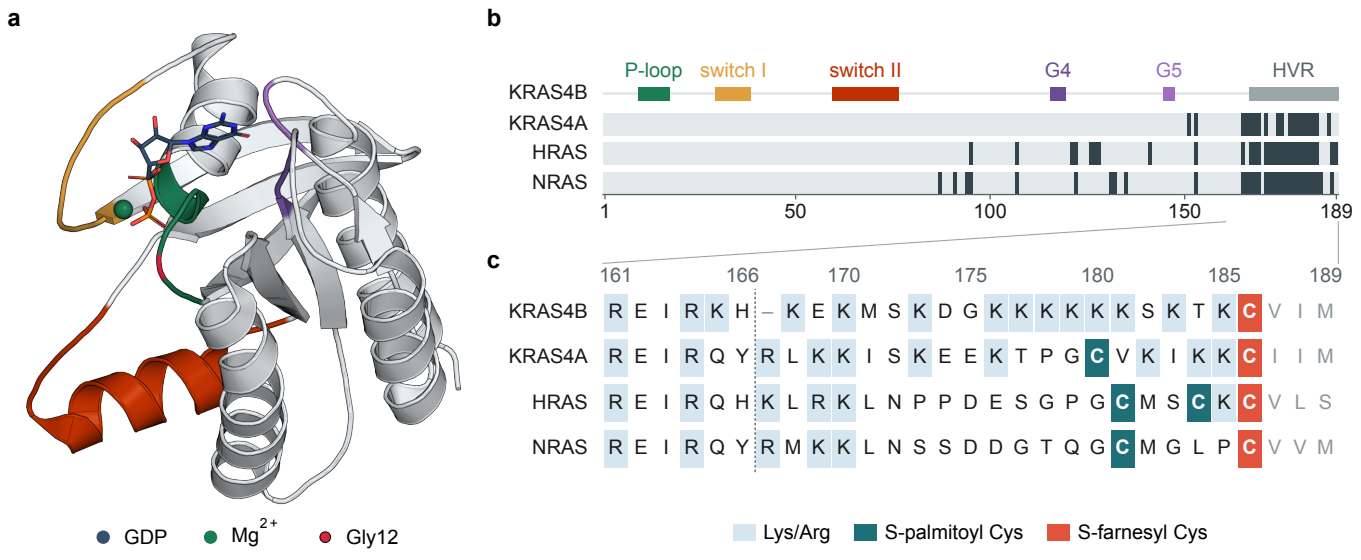

a
GDP
Mg 2+
Gly12
b
P-loop
switch I
switch II
G4
G5
HVR
KRAS4B
KRAS4A
HRAS
NRAS
1
50
100
150
189
c
161
166
170
175
180
185
189
KRAS4B R E I R K H – K E K M S K D G K K K K K K S K T K C V I M
KRAS4A R E I R Q Y R L K K I S K E E K T P G C V K I K K C I I M
HRAS R E I R Q H K L R K L N P P D E S G P G C M S C K C V L S
NRAS R E I R Q Y R M K K L N S S D D G T Q G C M G L P C V V M
Lys/Arg
S-palmitoyl Cys
S-farnesyl Cys